

HITOYOSHI, Japan

WMO: 478240

Lat: 32.2181N

Lon: 130.7553E

Elev: 484

StdP: 14.44

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 26.1	(c) 28.0	(d) 17.5	(e) 13.5	(f) 39.8	(g) 20.1	(h) 15.3	(i) 38.7	(j) 14.0	(k) 42.2	(l) 12.3	(m) 42.8	(n) 1.9	(o) 60	(p) 0.339

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 15.6	(c) 92.5	(d) 77.4	(e) 90.7	(f) 76.7	(g) 88.7	(h) 75.9	(i) 79.3	(j) 87.6	(k) 78.3	(l) 86.1	(m) 77.6	(n) 84.9	(o) 5.7	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.3	144.7	82.3	76.4	140.3	81.5	75.6	136.3	80.8	43.3	87.8	42.3	86.4	41.5	85.1	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.1	(b) 10.9	(c) 9.2	(e) 21.8	(f) 95.3	(g) 2.6	(h) 1.5	(i) 20.0	(j) 96.3	(k) 18.5	(l) 97.2	(m) 17.1	(n) 98.1	(o) 15.2	(p) 99.1		
(a) 13.1	(b) 10.9	(c) 9.2	(e) 21.1	(f) 81.6	(g) 2.6	(h) 2.0	(i) 19.3	(j) 83.1	(k) 17.8	(l) 84.2	(m) 16.4	(n) 85.4	(o) 14.5	(p) 86.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.5	41.1	44.1	50.1	59.0	67.2	73.2	80.2	81.3	75.9	66.0	54.7	44.5
	DBStd	14.82	4.89	6.40	5.99	5.74	4.04	3.99	3.13	2.59	4.42	5.51	6.59	5.86
	HDD50	763	278	187	74	6	0	0	0	0	0	0	27	191
	HDD65	3004	740	584	462	191	24	1	0	0	0	53	314	635
	CDD50	4976	4	23	77	275	532	695	936	972	778	496	167	21
	CDD65	1741	0	0	0	10	91	245	471	507	329	84	4	0
	CDH74	13838	0	0	4	100	640	1438	4173	4732	2320	422	9	0
	CDH80	4775	0	0	0	8	107	350	1611	1895	748	56	0	0
Wind	WSAvg	3.0	3.0	3.2	3.4	3.4	3.3	3.0	3.1	3.4	2.9	2.6	2.4	2.7
Precipitation	PrecAvg	98.1	2.9	4.2	6.3	7.2	8.3	21.7	16.4	9.7	9.8	4.6	3.9	3.2
	PrecMax	161.4	7.2	8.0	15.9	14.4	14.6	37.8	41.5	30.2	20.9	13.9	8.3	7.5
	PrecMin	58.8	0.2	1.9	1.9	2.1	2.5	7.0	2.7	1.4	1.0	0.6	1.1	0.3
	PrecStd	19.8	1.4	1.9	2.8	3.0	3.4	8.1	9.4	5.7	5.7	3.0	1.9	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.1	68.0	73.3	81.0	86.4	88.7	94.4	94.8	91.6	84.9	75.2	64.4
		MCWB	51.8	58.4	58.1	62.7	66.1	73.0	78.1	78.5	76.5	72.4	65.0	56.8
	2%	DB	56.0	62.7	68.7	76.9	82.8	86.2	92.5	92.9	89.0	81.3	71.1	60.5
		MCWB	49.0	55.0	55.6	60.9	64.7	72.2	77.4	77.6	74.8	69.3	62.5	55.0
	5%	DB	53.3	59.0	64.8	73.7	80.4	84.1	90.7	91.2	87.0	78.6	68.1	57.4
		MCWB	47.2	52.1	54.2	59.5	64.2	71.2	76.9	77.1	74.1	67.5	60.9	52.2
	10%	DB	50.5	55.4	61.4	70.6	77.7	81.7	88.5	89.2	84.7	75.9	65.3	54.5
		MCWB	45.0	48.7	52.9	59.2	63.7	70.6	76.5	76.7	73.4	66.4	59.9	49.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.8	61.0	62.5	69.3	73.4	77.4	80.9	80.9	79.0	75.4	68.4	59.9
		MCDB	57.7	64.5	65.6	72.7	78.2	83.1	90.6	90.3	86.7	80.8	71.8	61.8
	2%	WB	51.9	57.3	59.6	66.5	71.2	76.2	79.2	79.6	77.5	73.0	65.3	56.2
		MCDB	53.6	61.2	64.0	70.3	75.1	81.9	87.6	88.3	84.2	77.7	68.0	58.6
	5%	WB	48.5	53.6	57.5	63.7	69.2	75.1	78.2	78.7	76.4	71.0	63.1	53.6
		MCDB	51.6	57.3	62.3	69.0	73.6	80.2	86.0	86.8	82.6	75.6	66.4	56.4
	10%	WB	45.9	50.3	54.8	61.5	67.3	73.9	77.4	77.9	75.3	68.7	60.8	50.7
		MCDB	49.4	53.8	59.6	66.9	72.6	78.1	84.7	85.6	81.2	73.1	64.1	53.6
Mean Daily Temperature Range	5% DB	MDBR	16.9	18.5	19.6	20.8	19.9	14.5	14.9	15.6	16.5	18.5	17.8	16.9
		MCDBR	21.1	23.1	25.6	26.7	25.8	19.6	19.2	19.0	19.0	21.1	20.0	19.3
		MCWBR	13.8	14.3	14.1	12.5	10.7	7.4	6.7	6.5	7.1	9.7	11.2	12.6
	5% WB	MCDBR	16.6	17.4	18.2	16.9	15.1	12.5	16.2	16.0	15.0	14.5	13.7	15.2
		MCWBR	13.0	13.0	12.7	10.3	8.9	6.2	6.4	6.2	6.3	7.6	9.0	11.6
Clear-Sky Solar Irradiance	taub	0.413	0.451	0.509	0.529	0.539	0.543	0.536	0.520	0.497	0.453	0.415	0.403	
	taud	2.117	2.047	1.897	1.903	1.945	2.019	2.078	2.115	2.141	2.182	2.207	2.163	
	Ebn at Noon	247	250	245	246	245	243	244	246	246	247	246	243	
	Edh at Noon	41	48	60	62	60	56	52	49	46	42	37	37	
All-Sky Solar Radiation	RadAvg	783	985	1299	1519	1594	1294	1601	1641	1370	1154	873	727	
	RadStd	75	86	85	147	179	210	200	136	150	126	88	59	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page